**INFORMATION COMMUNICATION TECHNOLOGY**

**Object-Oriented Programming in C++**

**Module 4**

**CAT 3**

**Time:** 1 hour 30 minutes | **Total:** 40 marks

**Candidate Name: _____**

**Admission Number: _____ Date: _____**

**Instructions**

1. Answer ALL questions.
2. Write your name and admission number in the spaces provided above.
3. Write all code neatly and use proper C++ syntax.

**QUESTIONS**

1. Define **object-oriented programming** and state what an object bundles together. (2 marks)
2. Explain the difference between a **class** and an **object**, using a real-world example. (3 marks)
3. Compare **procedural programming** and **object-oriented programming** in terms of focus, handling of data, and reuse. (3 marks)
4. Explain the OOP concepts of **encapsulation** and **polymorphism**, giving one example of each. (4 marks)

5. State **two advantages** and **two disadvantages** of object-oriented programming. (4 marks)
6. State the purpose of each of the following parts of a C++ program: (4 marks)

   a) #include <iostream>

   b) using namespace std;

   c) int main()

   d) return 0;
7. The program below contains errors. Identify **four** errors and write the corrected statements. (4 marks)

```
#include <iostream>
using namespace std

int main() {
    int age;
    cin << age;
    cout >> "Age is " >> age;
    return 0;
}
```

17. a) Describe the three C++ access specifiers: **public**, **private** and **protected**. (3 marks)

    b) A programmer declares the balance attribute of a BankAccount class as **public**. Evaluate this decision and recommend a better approach. (3 marks)

18. a) Write a C++ class named Book with a string attribute title, an int attribute price, and a method display() that prints both values. All members should be public. (3 marks)

    b) Write the statements in main() that create an object of Book, assign values to its attributes, and call the display() method. (2 marks)

19. Write a complete C++ program that defines a class Rectangle with a method calculateArea(int length, int width) that returns the area. In main(), create an object, call

the method with values of your choice, store the result in a variable, and display it. (5 marks)